AMOD-N passive RF and analog blocks

Jason Gallicchio and Claude, rev A, 2026-10-25 github.com/gallicchio/AMOD-and-RFMOD

FOOT

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n9 Oscillator
Pmod 3.3 V
slide up: on
TUNE +-2 V
OUT -22 dBm

12 to 9 MHz

-2V +2V

n1 Calibration
SHORT, OPEN,
LOAD at the pin
25 OHM: VSWR 2
THRU 115 ps
tiles measure
from their pads

github.com/gallicchio/
AMOD-and-RFMOD
handbook and generator

n3 Transformer splitter
0.5-400 MHz
-3.2 dB each way
Isolation 25 dB
a combiner too

Isolation 0 to -40 dB

500k 400M

n2 Resistive splitter
DC-6 GHz
-6 dB each way
no isolation
IN max +20 dBm

n7 Bias-T
DC 16 V max
100 mA max
RF+DC has DC:
mind its load

n6 Detector
IN < +10 dBm
OUT to a
HIGH-Z ADC
square law
below -20 dBm

n5 Low-pass
DC-1 GHz
-3 dB 1.3 GHz
-30 dB 1.9-5 G

0 to -40 dB

0.1G 6G

0 to -60 dB

10M 5.5G

n4 Stub
notch 2.45 GHz
and 7.35 GHz
50 ohm

n11 Transformer
3-800 MHz
OUT: 2x V,
4x the Z
1: 200, 2: 49.9

n10 Balun
OUT floats
0.4-500 MHz
0.25 W, 30 mA

n8 Attenuator
NEEDS THE PMOD'S 3.3 V
by hand: SPI off, bits on
FPGA: bits off, SPI on
IN max +23 dBm, +9.5 dBm
below 0.5 MHz

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AMOD-N: passive RF and analog blocks. IN is each
tile's top-left jack, OUT its bottom-right. No USB:
n8 and n9 run on the Pmod's 3.3 V when it is plugged
in; everything else needs nothing. See the handbook.

Sheet:		
File: amod_n.kicad_pcb		
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